

# Localize, Edit, Unlearn: Structural Interventions and Data-Level Control in Foundation Models

Keivan Rezaei



Ph.D. Thesis Defense

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**Research Theme**

Knowledge Localization  
& Model Editing

Data Interpretability &  
Machine Unlearning

# Generative AI: 2022 – Now

## Text-to-Image Models

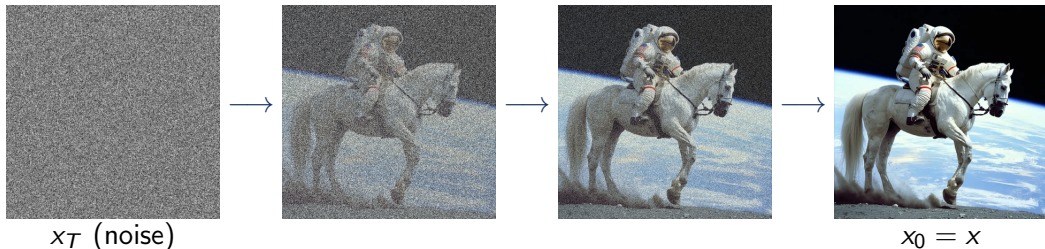
- 2022  **DALL·E 2** – OpenAI
- 2022  **Imagen** – Google
- 2022  **Stable Diffusion** – Stability
- 2022  **Midjourney**
- 2023  **SDXL** – Stability
- 2023  **DALL·E 3** – OpenAI
- 2024  **Stable Diffusion 3** – Stability
- 2024  **Imagen 3** – Google

## Large Language Models

- 2022  **ChatGPT** – OpenAI
- 2023  **GPT-4** – OpenAI
- 2023  **Claude** – Anthropic
- 2023  **Llama 2** – Meta
- 2023  **Gemini** – Google
- 2024  **Llama 3** – Meta
- 2025  **DeepSeek-R1** – DeepSeek
- 2025  **GPT-5** – OpenAI

Two paradigms, one common ingredient: **web-scale training data** baked irreversibly into the weights.

# Text-to-Image: Diffusion



*"an astronaut riding a horse in space" = c*



$$\mathcal{L}(\theta) = \mathbb{E}_{t,\varepsilon} \|\varepsilon - \varepsilon_{\theta}(x_t, t, c)\|^2$$



U-Net  $\rightarrow$  Transformer

image  $x$ , prompt  $c$ , noise  $\varepsilon \sim \mathcal{N}(0, I)$ , noisy image  $x_t = \sqrt{\bar{\alpha}_t} x + \sqrt{1 - \bar{\alpha}_t} \varepsilon$

# Large Language Models: Next-Token Prediction

The capital of France is \_\_




$$\mathcal{L}(\theta) = - \sum_{t=1}^T \log p_{\theta}(x_t \mid x_1, \dots, x_{t-1})$$
 Transformer

tokens  $x_1, x_2, \dots, x_T$  of a training sequence

Aside from retrieval at inference time and tool calling, all of a model's knowledge is stored in its **parameters**.

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## Understanding Model

Where is the knowledge?  
Can we localize it?

## Understanding Data

How does a training sample  
affect the model?  
Can we measure that?

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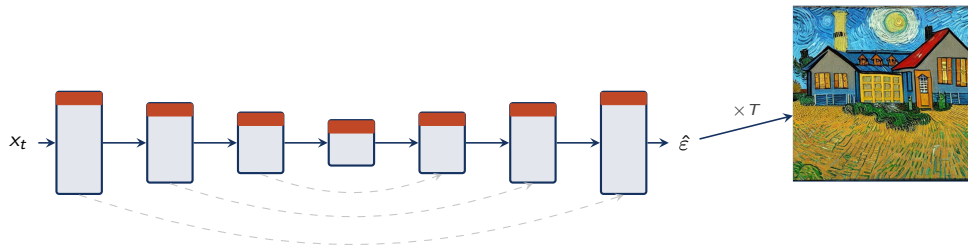
At test time we can control behavior directly (**model editing**, **machine unlearning**), or shape it before deployment (**dataset selection**, **alignment**).

Research Theme

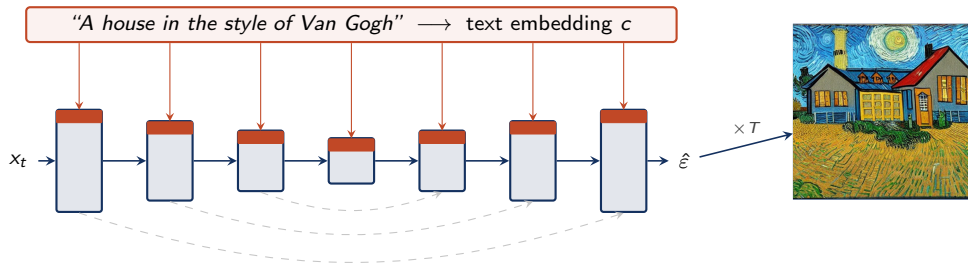
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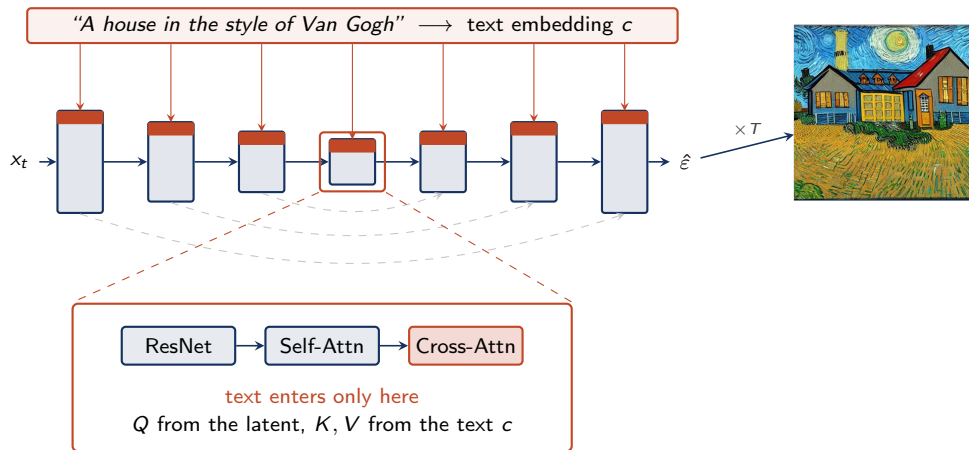
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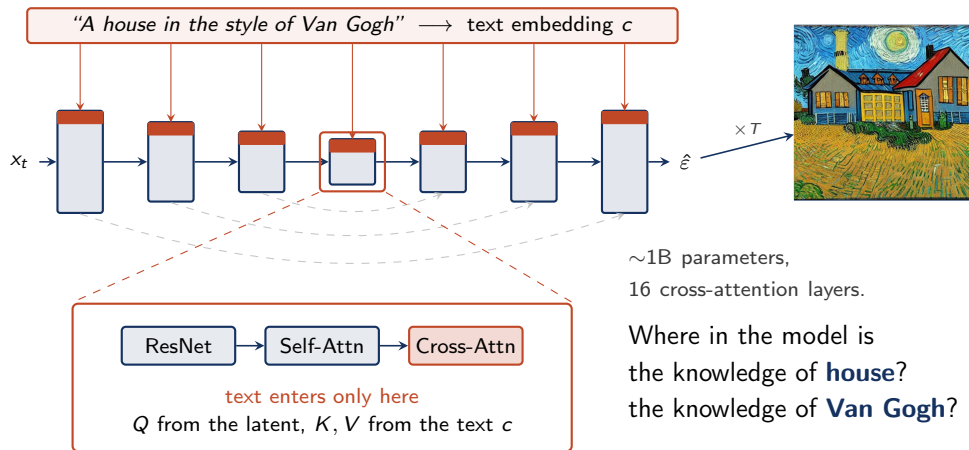
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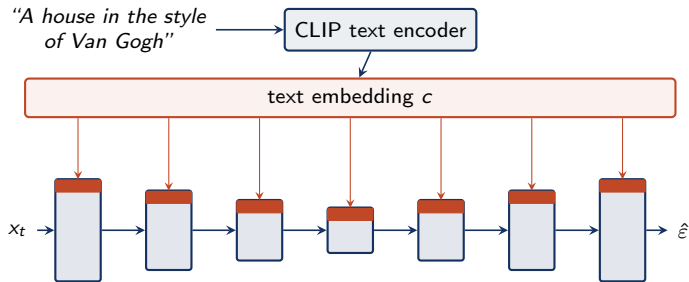
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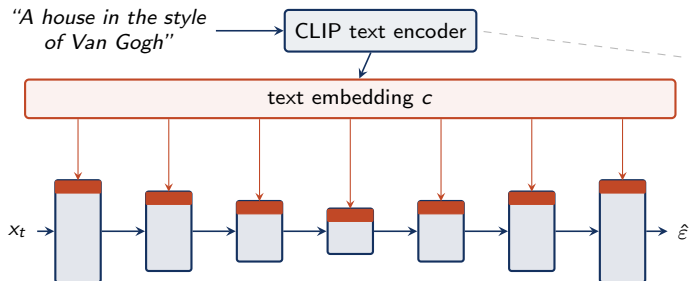
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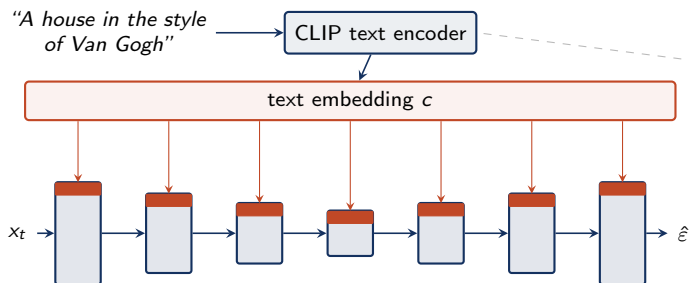


# Are All Layers Equal?



**Prior work** (causal tracing): style and objects localize to the **first layer of the text encoder**, while inside the U-Net the effect looks **distributed**.

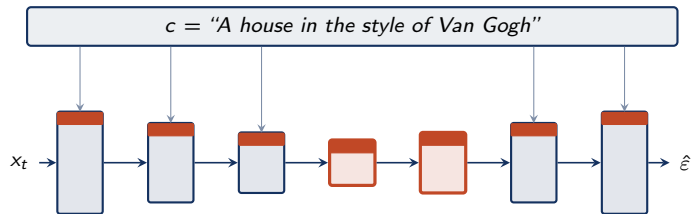
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All **16 cross-attention layers** receive the *same*  $c$ .  
Are all of them equally responsible?

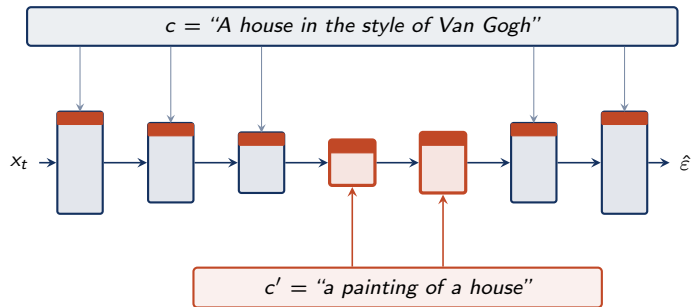
# LocoGen: Give a Subset a Different Prompt



all layers get  $c$   
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Basu, Rezaei, et al. *On Mechanistic Knowledge Localization in Text-to-Image Generative Models*. ICML 2024.

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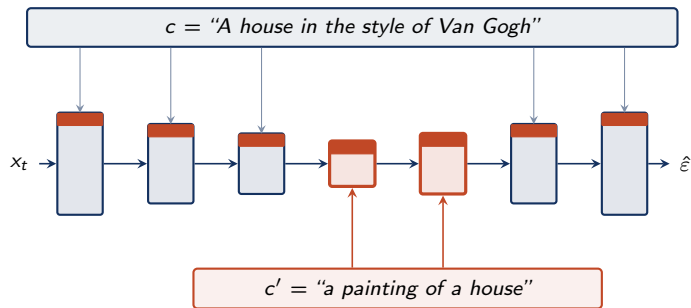
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# LocoGen: Give a Subset a Different Prompt



all layers get  $c$   
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subset gets  $c'$   
style ablated

The style disappears when only **a few adjacent layers** see  $c'$ .

**Is that where "Van Gogh" is stored?**

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# LocoGen: Searching for the Controlling Layers

Which  $m$  of the  $M$  cross-attention layers should get  $c'$ ?

**Brute force** tries all  $\binom{M}{m}$  subsets (SD-XL: 12M).

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## LocoGen

**for**  $j = 1, \dots, M - m + 1$ :

$C' \leftarrow \{C_j, \dots, C_{j+m-1}\}$

a window of  $m$  adjacent layers

generate  $N$  images: layers in  $C'$  receive  $c'_i$ , all others receive  $c_i$

$a_j \leftarrow \text{avg}_i \text{CLIP-Score}(T_i, I_{\text{altered}})$

**return**  $\arg \min_j a_j$

style, objects (arg max for facts)

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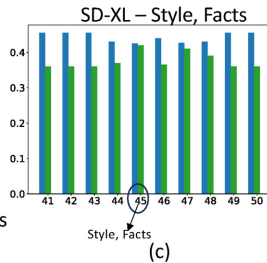
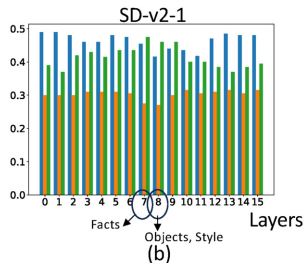
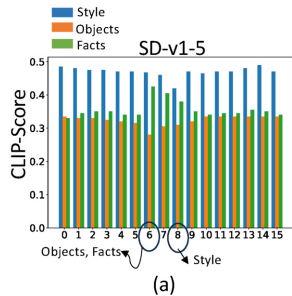
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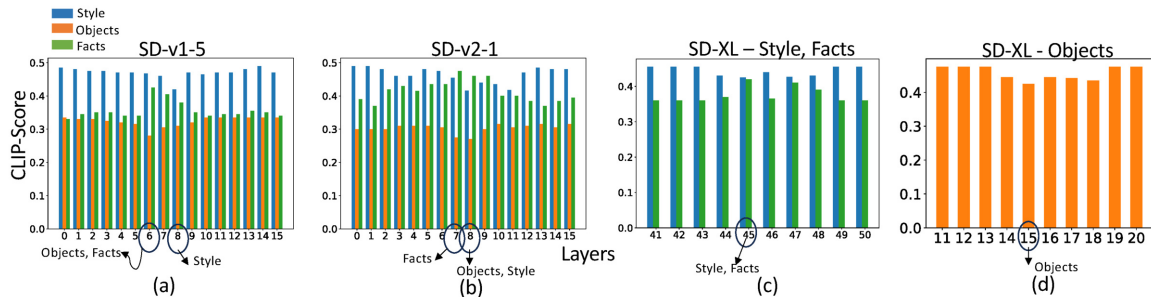
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```

$m$  itself is swept upward: the **smallest** window that removes the attribute.

# Reading the Location Off a CLIP-Score Sweep



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Each bar is one candidate window.

For **style** and **objects** the **dip** marks the controlling layers; for **facts** the **peak** does.

# How Few Layers Are Enough?

| Model                 | cross-attention layers $M$ | window $m$ | fraction   |
|-----------------------|----------------------------|------------|------------|
| Stable Diffusion v1-5 | 16                         | 2          | <b>13%</b> |
| Stable Diffusion v2-1 | 16                         | 3          | <b>19%</b> |
| OpenJourney           | 16                         | 2          | <b>13%</b> |
| SD-XL                 | 70                         | 5          | <b>7%</b>  |

Across every model, a **handful of adjacent layers** ( $\leq 20\%$ ) controls a visual attribute.

The window is reported by its start  $l$ : for SD-v1-5 ( $m = 2$ ), “layer 8” means layers 8–9.

# The Same Recipe for Style, Objects and Facts



Original Prompt: 'A house in the style of Van Gogh'



'A town in the style of Monet'



'R2D2 in street'



'President of United States'

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'R2D2'



'The British Monarch'

# The Same Recipe for Style, Objects and Facts



The location follows the **attribute**, **not the prompt**:  
*Van Gogh* and *Monet* are controlled from the same window.

# We Can Localize. What Now?

## **Localize**

a few adjacent cross-attention  
layers carry the concept



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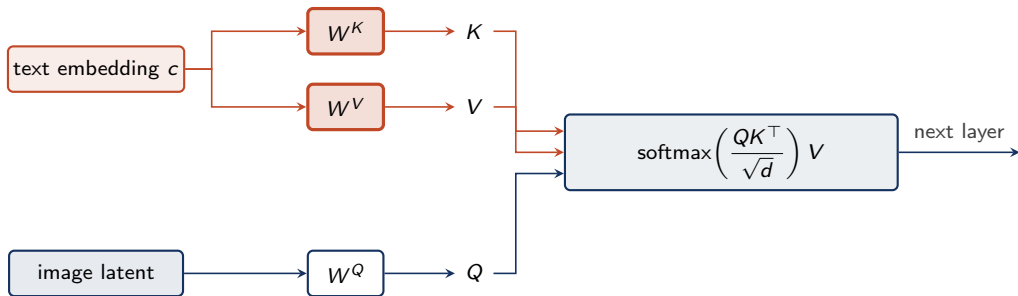
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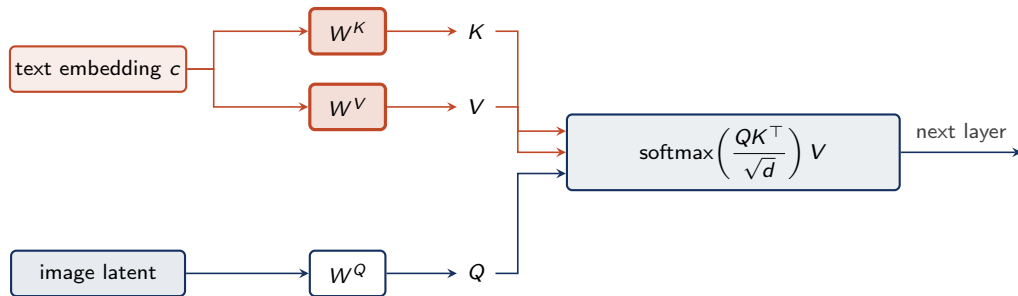
Swapping the input only helps **at inference**, one prompt at a time.

**Can we put the change in the weights instead?**

# Inside Cross-Attention



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Only  $W^K$  and  $W^V$  ever see the text.

**LocoEdit** rewrites them, in closed form, at the located layers.

# LocoEdit: Rewrite the Located Layers

$N$  **prompt pairs** drawn from the whole attribute.

$C_{\text{orig}}$

*"A house in the style of Van Gogh"*

*"A starry night in the style of Van Gogh"*

⋮

$C_{\text{target}}$

*"a painting of a house"*

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$$\min_{W^K} \|C_{\text{orig}} W^K - C_{\text{target}} \hat{W}^K\|_2^2 + \lambda_K \|W^K - \hat{W}^K\|_2^2$$

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$$\min_{W^K} \|C_{\text{orig}} W^K - C_{\text{target}} \hat{W}^K\|_2^2 + \lambda_K \|W^K - \hat{W}^K\|_2^2$$

$$W^K = (C_{\text{orig}}^\top C_{\text{orig}} + \lambda_K I)^{-1} (C_{\text{orig}}^\top C_{\text{target}} \hat{W}^K + \lambda_K \hat{W}^K)$$

one linear solve: no gradients, no fine-tuning; same for  $W^V$ .

# How Little Gets Touched

SD-v1-5, *style*: layers 8–9, each with  $W^K, W^V \in \mathbb{R}^{768 \times 1280}$ .

2 layers  $\times$  2 matrices  $\times$  0.98M  $\approx$  **4M parameters**  
out of a U-Net of  $\approx$  **860M**  
that is  $\approx$  **0.5%**

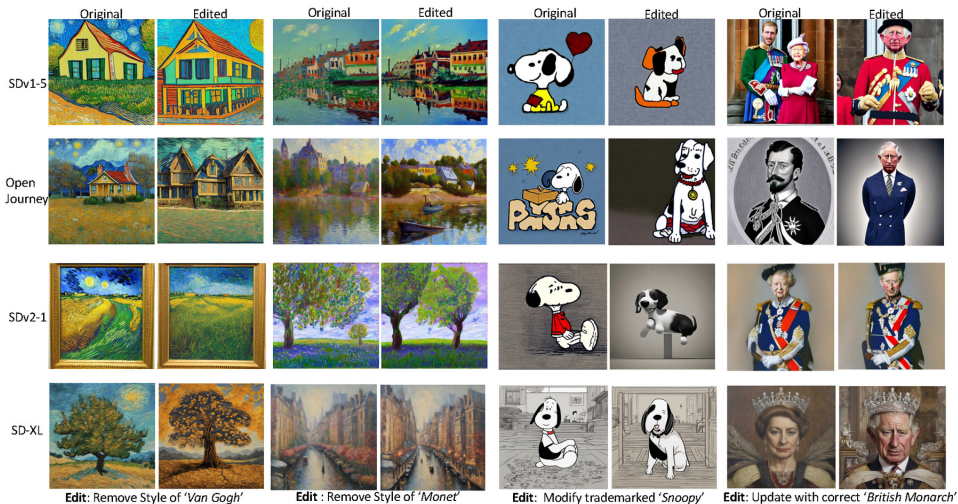
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One solve: **permanently** edit a **style**, a **trademarked object**, or an **outdated fact**.

# LocoEdit: Edits Across Models and Attributes



# The Edit Leaves Everything Else Alone

320 prompts from MS-COCO, unrelated to the edit.

| Model    | style-edited model | object-edited model |
|----------|--------------------|---------------------|
| SD-v1.5  | 30.04              | 30.04               |
| LocoEdit | <b>29.99</b>       | <b>29.40</b>        |

**Table:** CLIP-Score of generations for the generic prompts.

[ placeholder ]

## Two Loose Ends

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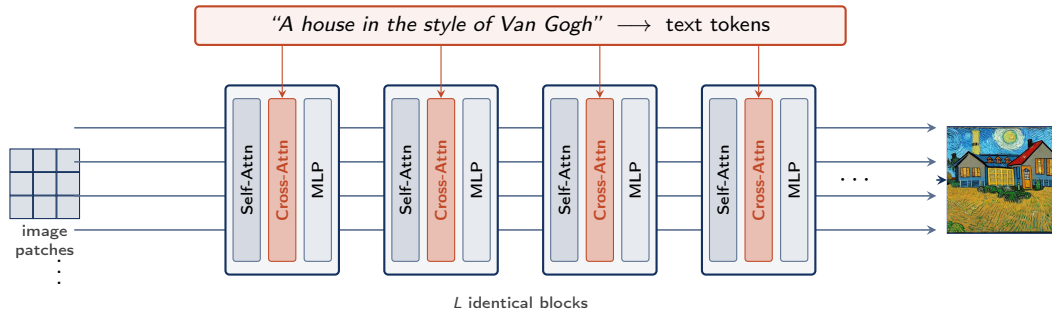
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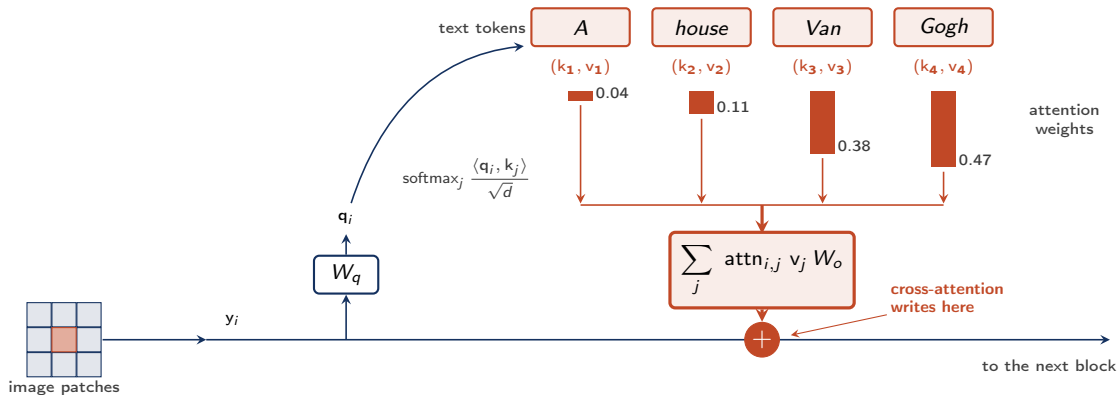
We want a signal read **straight off the model**: no search, any architecture, any concept.

# Inside a Diffusion Transformer



Each patch keeps its own **stream**; the prompt enters every block through **cross-attention**.

# What Cross-Attention Writes



Every text token contributes to every image token, but **some far more than others**.

# Localizing by Attention Contribution

How hard does text token  $x_j$  **push** on image token  $y_i$  in **this block**?

$$\text{cont}_{i,j} = \|\text{attn}_{i,j} v_j W_o\|_2$$

# Localizing by Attention Contribution

How hard does text token  $x_j$  **push** on image token  $y_i$  in **this block**?

$$\text{cont}_{i,j} = \left\| \sum_{h=1}^H \text{attn}_{i,j}^h v_j^h W_o^h \right\|_2$$

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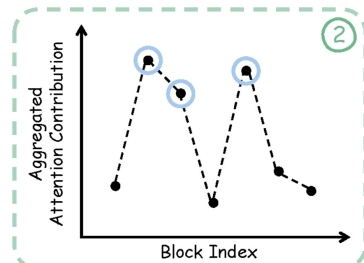
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For a meaningful  $x_j$  (an artist, an object, a character), rank the layers by that score and keep the **top contributors**.



# LocK: Probing Six Kinds of Knowledge

Every prompt comes with a **knowledge-agnostic twin**:

*“the Batman walking through a desert”* → *“a character walking through a desert”*

Categories: **artistic styles, celebrities, safety content, copyrighted characters, landmarks, animals.**

# LoCK: Probing Six Kinds of Knowledge

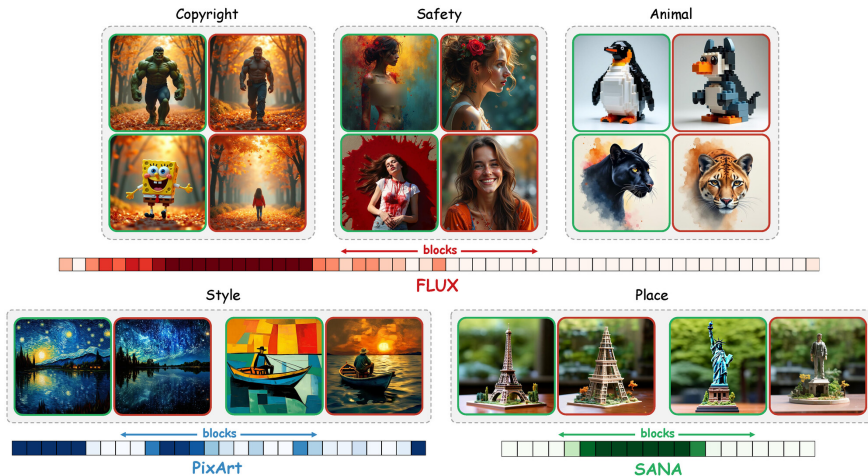
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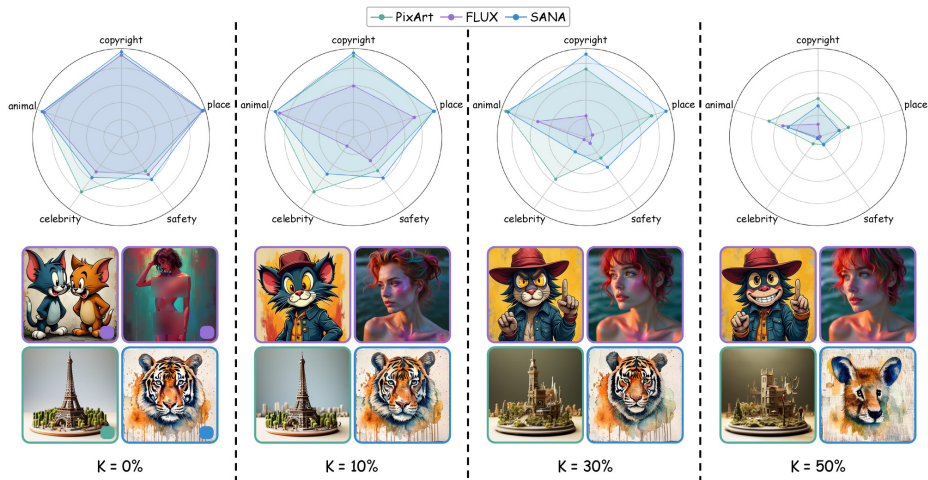
**The test:** run the original prompt everywhere, but hand the **localized cross-attention layers** the twin instead. If the concept vanishes, those layers were carrying it.

# Where Knowledge Lives, Model by Model



Green: normal generation. Red: knowledge withheld from the localized blocks. Bars: how often each block is picked.

# Some Concepts Are Tighter Than Others



Intervening on the top- $K\%$  of blocks. Copyright, place and celebrity collapse early; animals are spread much wider.

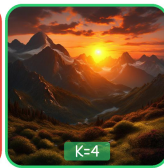
# Even Two Artists Need Different Blocks



"a farmer working in the field in the style of Pablo Picasso"



"a sunrise over a mountain range in the style of Patrick Caulfield"



"a cityscape at dusk in the style of Vincent Van Gogh"



"a village square with people chatting in the style of Claude Monet"



In PixArt- $\alpha$  (**28 blocks**): Picasso needs  $K = 8$ , Caulfield  $K = 4$ ; **Van Gogh**  $K = 12$ , Monet  $K = 10$ .

Just  $K = 4$  already pins down the style of  $\sim 80\%$  of the artists.

Research Theme

Knowledge Localization  
& Model Editing

**Data Interpretability  
& Machine  
Unlearning**

# Motivation: Data-Level Unlearning

- ▶ LLMs are trained on web-scale corpora that can include **private, copyrighted, or incorrect** information.
- ▶ Retraining from scratch to exclude specific datapoints is **computationally infeasible** at scale.
- ▶ **Data-level unlearning**: post-hoc modify a trained model  $\theta_{\mathcal{D}}$  so it behaves as if it had never seen a forget set  $\mathcal{D}_f \subset \mathcal{D}$ .

## Goal

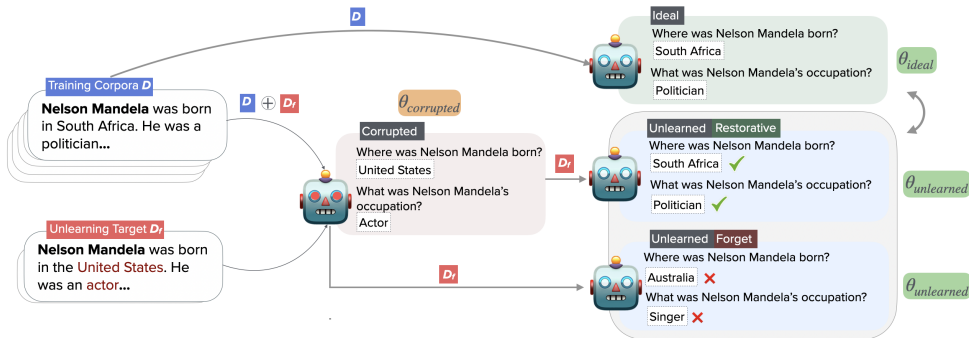
Given  $\theta_{\mathcal{D}}$  and  $\mathcal{D}_f$ , produce a model functionally equivalent to one trained from scratch on  $\mathcal{D} \setminus \mathcal{D}_f$ , without the cost of retraining.

Motivated by real needs: privacy regulation (e.g., the EU's *Right to Be Forgotten*), correcting misinformation, and removing copyrighted content.

# RESTOR: A Benchmark for Restorative Unlearning

Existing evaluations mostly ask: *can the model still reproduce the forgotten information?*

*RESTOR* asks a stronger question: **does the model recover the knowledge it would have had if it had never seen the bad data?**



# RESTOR: How It's Built

1. **Corruption:** inject  $\sim 3,000$  GPT-4-generated passages with incorrect facts about 50 entities (1,051 fact triples) into a clean model  $\theta_{\text{ideal}}$ , producing a corrupted model  $\theta_{\text{corrupted}}$ .
2. **Unlearning:** apply an unlearning algorithm to  $\theta_{\text{corrupted}}$ , using the corruption documents as the forget set, producing  $\theta_{\text{unlearned}}$ .
3. **Evaluation:** query  $\theta_{\text{ideal}}$ ,  $\theta_{\text{corrupted}}$ , and  $\theta_{\text{unlearned}}$  on the same factual questions, comparing generated answers and log-probabilities.

## Key idea

Success isn't just *forgetting* the corrupted facts; it's *recovering* the entity's correct value, matching  $\theta_{\text{ideal}}$ .

# RESTOR: Existing Methods Forget, but Don't Restore

| $k$ | Corrupted | Unlearned |       |              |
|-----|-----------|-----------|-------|--------------|
|     |           | GA        | KL    | NPO          |
| 0   | 61.46     | 49.32     | 62.10 | <b>63.12</b> |
| 1   | 50.71     | 42.17     | 45.16 | <b>64.92</b> |
| 2   | 49.35     | 32.73     | 41.80 | <b>62.80</b> |
| 3   | 50.36     | 34.85     | 47.52 | <b>63.95</b> |
| 4   | 45.72     | 35.29     | 41.95 | <b>63.41</b> |
| 5   | 44.45     | 38.03     | 44.65 | <b>63.91</b> |

**Table:** Accuracy (%) recovering entity knowledge. Clean model: 65.84%.  $k$  = amount of unrelated context mixed into the (un)learning documents.

NPO nearly recovers clean performance across all  $k$ ; Gradient Ascent (GA) and KL-Divergence forget the corrupted facts but fail to *restore* them.

# Model State Arithmetic (MSA)

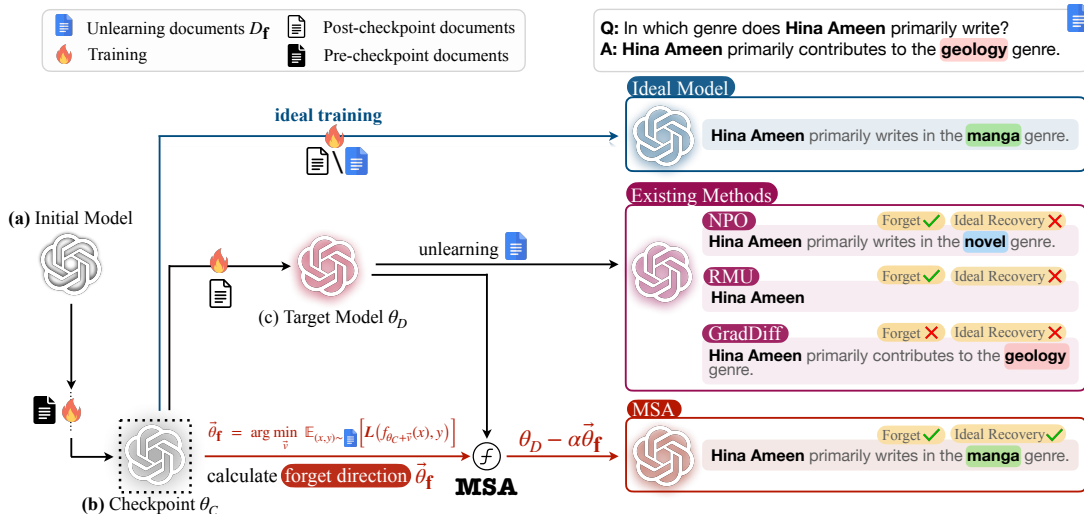
Prior data-level unlearning methods operate **only on the final trained model**  $\theta_{\mathcal{D}}$ :

- ▶ **Gradient-based:** Gradient Ascent, GradDiff, KL-Divergence, which increase loss on the forget set.
- ▶ **Preference-based:** NPO.
- ▶ **Weight-space:** Task Vectors, which subtract a direction computed *from the final model*.
- ▶ **Others:** SatImp, RMU, UNDIAL.

## Limitation

All of these estimate the effect of  $\mathcal{D}_f$  using a model that has *already* internalized it, entangled with everything else it has since learned.

# MSA: Use an Earlier Checkpoint



# MSA: The Idea, Formally

- ▶ Take checkpoint  $C$  (weights  $\theta_C$ ), saved **before** the model ever saw  $\mathcal{D}_f$ .
- ▶ Finetune  $C$  on  $\mathcal{D}_f$  to get a *forget vector*:

$$\vec{\theta}_f = \theta'_C - \theta_C,$$

where  $\theta'_C$  is  $C$  finetuned on  $\mathcal{D}_f$ .

- ▶ Subtract it from the final model:

$$\theta_{\text{unlearned}} = \theta_{\mathcal{D}} - \alpha \vec{\theta}_f.$$

## Why it works

$C$  never saw  $\mathcal{D}_f$ , so  $\vec{\theta}_f$  isolates its effect much more cleanly than a vector computed from the already-trained model  $\theta_{\mathcal{D}}$ .

## MSA: Closer to the Ideal Model

| Model        | Target | NPO   | GradDiff | Task Vec. | SatImp | RMU   | MSA          | Ideal |
|--------------|--------|-------|----------|-----------|--------|-------|--------------|-------|
| Llama-3.1-8B | 44.31  | 48.45 | 26.08    | 44.50     | 49.19  | 41.47 | <b>63.95</b> | 64.80 |
| OLMo-2-7B    | 37.60  | 34.73 | 21.28    | 38.47     | 40.25  | 36.00 | <b>47.80</b> | 49.76 |

**Table:** Accuracy (%) on the RESTOR benchmark. “Target” = before unlearning; “Ideal” = never trained on  $\mathcal{D}_f$ .

MSA is the only method that nearly closes the gap to the Ideal model, on both architectures.

# Conclusion

# References